

Sovereign Ingestion & Email Soft-Token Architecture

Bridging Microsoft Dynamics NAV / Navision to Stride Sovereign Cloud via Ephemeral Dual-Factor Soft Tokens & Zero-Trust Tokenization

EXECUTIVE MANDATE & AUDIT SUMMARY

Deposit-Taking SACCOs in Kenya running on-premise ERPs (principally Microsoft Dynamics NAV / Business Central) require seamless, daily or monthly data synchronization into StrideAnalytics for automated SASRA Form 1 & 4 returns and stress testing. This whitepaper establishes Stride's **Air-Gapped Soft-Token Upload Protocol** — delivering banking-grade security with zero IT friction, zero direct port exposure, and 100% ODPC statutory data compliance.

1. The Operational Challenge: Navision Data Extraction Without Opening Firewall Ports

Tier-1 SACCO internal core banking servers hosting Microsoft Dynamics NAV (Navision) are strictly isolated behind corporate DMZs. Exposing database ports (e.g. MS-SQL 1433) or direct Web Services (SOAP/OData) directly to the public internet creates intolerable cyber risk and violates CBK/SASRA cybersecurity guidelines.

Stride's Solution: Stride utilizes an *Outbound-Push & Ephemeral Token Model*. Neither the SACCO nor Stride requires inbound firewall pinholes. Authorized SACCO Finance or IT Officers export standardized XML/CSV general ledgers and trial balances from Navision, and transmit them via Stride's secure, single-use email soft-token gateway.

2. The 4-Step Email Soft-Token Lifecycle (End-to-End Workflow)

The authentication workflow eliminates permanently stored static passwords for data uploads, mitigating credential theft and unauthorized batch modifications:

Step	Mechanism	Security Control & Safeguard
01. Upload Request Trigger	Authorized Finance Officer enters registered email on tenant.strideanalytics.co.ke/ingest.	Whitelisted against SACCO Board authorized signatory database. Rate-limited to prevent enumeration.
02. Dual-Hash Token Dispatch	Stride Auth-Engine issues an encrypted, single-use 6-digit TOTP plus an embedded Magic Link to corporate email.	Valid for exactly 15 minutes . Delivered via end-to-end encrypted SMTP TLS with SPF/DKIM verification.
03. Secure Vault Session	Officer clicks the magic link or enters the 6-digit soft token. An encrypted memory buffer opens.	Session is client-side pinned. Automatic revocation upon upload completion or window expiry.
04. Client-Side ODPC Tokenization	Navision CSV/Excel file is processed. Member National IDs and phone numbers are SHA-256 salted before upload.	Zero raw personal identifiers reach the cloud. Merkle root proof returned for SACCO audit files.

3. Microsoft Dynamics NAV Integration & 'Running Balance' Alignment

Following feedback from institutional regulators and executive SACCO management (governed under SASRA Prudential Directives), Stride ingests financial trial balance balances as **Absolute Point-in-Time Running Balances** rather than arbitrary summation flows:

- **Balance Sheet Accounts (Loan Book & Member Deposits):** Extracted from Navision General Ledger closing positions as at December 31st or month-end. Stride's engine recognizes these as cumulative stock reserves.
- **Income & Expense Accounts (Net Influx Δ Flows):** Extracted as period transactions, allowing boardrooms to analyze both total asset size and YoY growth delta simultaneously.
- **Pre-Built Navision XML Port / Dataport:** Stride provides an official 1-click XML Port for Dynamics NAV 2013, 2016, 2018, and Business Central that packages required trial balance columns in under 30 seconds.

4. Institutional Security Evaluation: Stride vs Legacy Methods

Security Dimension	Traditional FTP / Open Port	Stride Sovereign Soft-Token Gateway
Attack Surface	High (Open ports 1433/21 exposed to public internet).	Zero Exposure (Outbound HTTPS 443 only; no listening ports).
Credential Vulnerability	Static passwords stored in plaintext configuration files.	Ephemeral TOTP (Tokens self-destruct in 15 mins).
ODPC PII Compliance	Raw member ID cards and phone numbers uploaded unencrypted.	Client-Side Dynamic Masking (SHA-256 salted tokens).
Audit Trail & Merkle Proof	Basic server access log; easily manipulated.	Cryptographic Ledger signed with time-stamped hash.

5. Regulatory Compliance & Statutory Attestation

StrideAnalytics operates under full statutory alignment with Kenyan laws governing financial and data institutions:

- **Sacco Societies Act (Cap 490B) Section 24:** Automates filing integrity for Form 1 (Capital Adequacy) and Form 4 (Liquidity Shield) directly from verified Navision balances.
- **Kenya Data Protection Act (ODPC 2019):** Stride ensures that in the event of an institutional inspection, all data custody is mathematically demonstratable as anonymized.
- **CBK Prudential Guidelines & Cybersecurity Framework:** Multi-tenant isolation guarantees zero data leakage between peer SACCO entities.

FOR INSTITUTIONAL PILOTS & NAVISION INTEGRATION KITS

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Kenya Data Protection Act (ODPC 2019) Compliant Architecture • SASRA Tier-1 Ready